

CarbonLens Storage Studio: Strategic Analysis Across PhD Worthiness, Industry Adoption, Technical Gaps, and Commercial Pricing

TL;DR

- **Yes — CarbonLens is a defensible PhD topic, but only if framed as ML-augmented CCS workflows + UQ benchmarking, not as a "browser app" PhD.** UTP's PETRONAS-aligned positioning is ideal, and Malaysia's CCUS Act 2025 (gazetted 1 October 2025, first permit granted to PCCSV at Duyong on 10 October 2025) creates a once-in-a-decade window to anchor research to a real regulatory regime. Precedent is strong: CMG itself was founded out of Khalid Aziz's University of Calgary research (1978); CMG posted CAD \$129.4 million revenue in FY2025 (+19% YoY, GlobeNewswire 22 May 2025) across roughly 600 customers in 60 countries (D&B company profile).
- **CarbonLens is pitchable as a screening / pre-FEED / permit-prep companion tool — NOT as a replacement for CMG GEM, SLB INTERSECT, or tNavigator at the FID stage.** Verified pricing benchmark: Petrel licenses cost USD \$30,000–\$50,000 per user per year (Open iT). CMG GEM and tNavigator pricing is bespoke and not publicly disclosed; tNavigator's full module bundle list-price reached USD \$3.75 million in a 2022 academic donation MoU with UCSI University. The realistic CarbonLens entry-point is USD \$15K–\$40K/user/year SaaS as a "pre-screening + permit-export" layer, with a credible path to USD \$80K–\$150K/year enterprise tier.
- **The single biggest technical gap is full 3D compositional multi-phase flow with coupled geochemistry and history matching.** CarbonLens's MARS-IFT + Span-Wagner + Duan-Sun + Mohr-Coulomb stack is correctly chosen for screening and uncertainty quantification, but US EPA Class VI Area of Review modeling and the EU CCS Directive 2009/31/EC Annex I Step 3 ("dynamic modelling of the CO₂ storage to reduce uncertainty and risk") both require dynamic 3D pressure-volume-saturation simulation that browser-only WebGL/WASM cannot deliver at field scale. Position CarbonLens as the *screening + UQ + permit-export* tool that *feeds* CMG GEM / INTERSECT — not a substitute.

Key Findings

1. PhD Research Worthiness — Strong, with caveats

Precedent. The single most important precedent is CMG itself: per Wikipedia's CMG company history, "The company began in 1978 as an effort to develop a simulator by Khalid Aziz of the University of Calgary's Chemical Engineering department, with a research grant from the government of Alberta." UTCHEM (chemical EOR simulator) was developed at UT Austin and remains the benchmark for chemical-flooding simulation 25+ years later (ScienceDirect). SINTEF's MRST and MRST-co2lab — the most directly comparable academic codebase to CarbonLens — emerged from the Applied Computational Science group at SINTEF Digital and has produced two Cambridge University Press monographs: (1) Lie, K.-A., *An Introduction to Reservoir Simulation Using MATLAB/GNU Octave: User Guide for the MATLAB Reservoir Simulation Toolbox (MRST)*, Cambridge University Press, 2019; and (2) Lie, K.-A. & Møyner, O., *Advanced Modeling with the MATLAB Reservoir Simulation Toolbox*, Cambridge University Press, November 2021

(SINTEF official MRST page). The "software-as-PhD" model is well-established in petroleum/reservoir engineering when paired with novel physics, validation, or ML contributions.

Novel research contributions CarbonLens can plausibly generate (PhD-grade):

1. **MARS IFT model validation paper** — benchmark MARS against published random-forest, XGBoost, ENN and Super Learner models. Recent published benchmarks include Nait Amar et al. (*Energy & Fuels* 2025, "Rigorous Explainable Artificial Intelligence Models for Predicting CO₂-Brine Interfacial Tension," Super Learner achieved RMSE = 0.7813 and $R^2 = 0.9953$ on a 2,616-point dataset) and the Bayesian-optimized random forest model (ACS Omega 2024, RMSE = 1.7705, MAPE = 2.0687%, $R^2 = 0.9729$). Target: *Fuel, International Journal of Greenhouse Gas Control* (2024 IF 6.19; Q1; Editor-in-Chief Prof. Samuel Krevor, Imperial College London; APC USD \$3,900).
2. **Browser-native UQ methodology paper** — P10/P50/P90 propagation through ML surrogates of EOS/solubility, compared with CMG CMOST. Target: *Computers & Geosciences, SPE Reservoir Evaluation & Engineering*.
3. **Permit-template formalization paper** — first formal cross-jurisdictional comparison of EPA Class VI / EU 2009/31/EC Annex I / Malaysian CCUS Act 2025 / OPGGS Act / NPD storage-permit technical schemas as a machine-readable ontology. Target: *International Journal of Greenhouse Gas Control* or *Energy Policy*.
4. **Benchmark study vs. industry simulators** — CarbonLens vs. CMG GEM / MRST-co2lab / TOUGH on canonical CCS test cases (Sleipner, Johansen, Utsira). Target: *Advances in Water Resources, Transport in Porous Media*.

Entrepreneurial/EngD route. The UK EngD is the cleanest entrepreneurial doctorate model: "An EngD (Engineering Doctorate) is a professional, industry-focused doctoral degree designed to prepare engineers for high-level technical and leadership roles. It sits at the same academic level as a PhD but has a different emphasis... work-based format" (IDCORE/University of Edinburgh). Heriot-Watt offers an **EngD Energy** (Dubai/Edinburgh). For CCS specifically, Edinburgh's SCCS group (the largest CCS research group in the UK, per energy.eng.ed.ac.uk) and Imperial College London are the strongest UK options. **At UTP itself**, the institutional structure favors entrepreneurial research: UTP holds Malaysia's MyRA **6-star rating** for research, development and commercialisation (the highest MoHE rating), is wholly owned by PETRONAS, and CCS/CO₂ Management is one of its declared niche areas. A UTP PhD by Research with explicit commercialization milestones, supplemented by an external EngD-style industrial collaboration (PCCSV, JOGMEC, TotalEnergies, or Mitsui — all already partnering on Duyong), is the realistic recommendation.

2. Industry Adoption & Pitchability

Malaysia / PETRONAS context is uniquely favorable. PETRONAS' Kasawari CCS reached FID in November 2022 (Petronas Carigali; SK316, 108 m water depth) and targets injection of 3.3 Mtpa with ~80 MT of cumulative storage over 25 years — positioned at FID as "the world largest offshore CCS upon its commencement" (Kasawari IPTC 2023 paper, OnePetro). Duyong received Malaysia's first Offshore Assessment Permit under the CCUS Act 2025 on 10 October 2025 (PETRONAS official release, 10 November 2025). The Malaysia Carbon Capture, Utilisation and Storage Agency (MyCCUS) is now the licensing authority. PETRONAS also signed JSAs with ADNOC + Storegga (Penyu basin, ≥5 Mtpa target by 2030) and with TotalEnergies + Mitsui for Duyong. The PETRONAS statement on the ADNOC/Storegga JSA explicitly mentions "geophysical and geomechanical modelling, reservoir simulation and containment research while exploring the application of advanced technologies, including artificial intelligence, to enhance storage capacity" (Argus Media, 20 August 2025). This is direct, named demand for the exact capability stack CarbonLens offers.

Global simulation-software market sizing. No CCUS-specific simulation-software market figure was found in primary sources; the umbrella simulation-software market is variously sized at **USD \$15.46 billion in 2026, projected to reach USD \$28.59 billion by 2031 at 13.08% CAGR** (Mordor Intelligence), USD \$19.95B (2024, MarketsandMarkets, 10.4% CAGR), and USD \$20.7B (2024, SkyQuest, 11.83% CAGR). The CCS-software slice is small (<1%) but is growing dramatically faster than the average given DOE 45Q tax credits, EU Industrial Carbon Management Strategy, and Asian CCS-hub demand. **Cloud/SaaS deployment is the fastest-growing segment** at 13.22% CAGR (Mordor), and per Fortune Business Insights "the cloud segment is projected to dominate the market with a share of 79.85% in 2026" — directly validating the browser-native thesis.

Competitive landscape. The CCS simulation stack is already crowded at the top end: CMG's Focus CCS™ and GEM™ ("the gold-standard simulator for CO₂ and H₂ storage processes," per cmgl.ca); SLB's INTERSECT (launched next-gen CO₂ storage modeling 2025) on Delfi cloud; tNavigator (Rock Flow Dynamics, per Groundwork Analytics "delivering simulation speeds that are often 5-10x faster than Eclipse on equivalent models using multi-core CPUs and GPU acceleration," with \$33.8M revenue in 2024 and 225 employees per Latka); Halliburton DecisionSpace + Resoptima (acquired June 2023; per Halliburton press release 5 June 2023, "To date, more than 130 active fields globally have benefited from Resoptima's technology"); MRST-co2lab (free, MATLAB-based, SINTEF). What is **not** crowded is the **screening / pre-FEED / permit-prep / UQ** segment — where a browser-native ML tool can credibly land. SimScale, the leading browser-native CAE comparable, closed €27M Series C (Insight Partners, January 2020) and a further €25M extension (October 2021, Draper Esprit + Insight, total ~€52M / ~\$60M cumulative). Per the SimScale Wikipedia entry citing the company's own September 2024 announcement, "In September 2024, the company announced that it had 600,000 registered users on its platform." This is the proof case that browser-native engineering simulation is a viable, fundable, large-market category.

Pain points (validated from sources). From Capterra/G2 reviews of Petrel and CMG and reservoir-engineering practitioner blogs: (a) license cost is the #1 complaint ("License cost is very high which does not include plugins cost," Capterra; "\$30,000 to \$50,000 per user" annually for Petrel, Open iT); (b) Windows-only / limited platform compatibility ("Petrel is primarily designed for Windows operating systems," G2); (c) computational burden — per Groundwork Analytics, "a full-field simulation model with a million cells can take hours to days for a single run, making iterative workflows (history matching, sensitivity analysis, optimization) time-consuming"; (d) vendor lock-in; (e) steep learning curve. CarbonLens's browser-native, zero-install, ML-accelerated value proposition directly attacks (a), (b), (c), and partially (e).

3. Technical Gap Analysis

EPA Class VI is the most demanding jurisdiction. The DOE/NETL/EPA "Rules and Tools Crosswalk" report (NRAP-TRS-I-001-2022; DOE/NETL-2022/3731; EPA-900-B-22-001; published 31 May 2022) catalogues **59 computational tools** for Class VI permitting. Per the report, "Reservoir simulation tools were the most frequently referenced tool type, with 16 separate responses provided. Other common tool types addressed seismic and geomechanical risks (7 responses provided) and geologic model development (7 responses provided)." Named tools include TOUGH (LBNL), STOMP (PNNL), NUFT (LLNL), FEHM (LANL), MRST (SINTEF), CMG (GEM/IMEX/STARS), MODFLOW (USGS), PHREEQC, Geochemist's Workbench, SGeMS, Open-IAM (NRAP), E4D, DREAM, SOSAT, EASiTool, and CO₂-SCREEN. Class VI demands include (i) Area of Review (AoR) modeling using 3D reservoir simulation, (ii) maps/cross-sections/geomechanical info on fractures/stress/rock strength (per Opportune's permit-process summary, "geo-mechanical information on fractures; and stress, ductility, rock strength, and in-situ fluid pressures within the confining zones"), (iii) baseline geochemical data of all USDWs, (iv) CO₂-rock-fluid compatibility analysis, and (v) static earth

model + dynamic 3D simulation. CarbonLens covers screening-level (i), partial (ii) via Mohr-Coulomb/MAIP, and (iv) via Duan-Sun + MARS IFT — but **NOT** full 3D compositional simulation, geochemical kinetics, or history matching.

EU CCS Directive 2009/31/EC Annex I prescribes a 3-step process: (1) data collection, (2) "building a three-dimensional model of the storage complex and its surroundings," (3) "dynamic modelling of the CO2 storage to reduce uncertainty and risk." The 2024 revised Guidance Documents (DNV + EU Commission) per Haavind, "sharpened the focus on... more detailed information on geomechanical characterization of the storage and a specific guidance on how to evaluate risks related to leakage via legacy wells." CarbonLens's geomechanics module (Mohr-Coulomb, MAIP) is well-aligned with Step 2 but is screening-grade for Step 3.

Malaysia CCUS Act 2025 / Offshore Permit and Licensing Regulations 2025: requires submission detailing "the area intended for offshore geological assessment and the intended methods and techniques for assessment" (Pinsent Masons/Skrine, PFI Yearbook, 19 December 2025). Per the same source, "Each offshore assessment permit application will cost M\$80,000 (US\$19,450)... An offshore assessment permit grants the right to undertake offshore geological assessment during the permit period within three years of obtaining the permit... Failure to comply with the conditions of an offshore assessment permit ... can result in a fine of up to M\$500,000." Per Azmi & Associates (Lexology), fees are "(i) an offshore assessment permit (RM80,000.00 per application), (ii) an extension of offshore assessment permit (RM80,000.00 per application), (iii) an offshore storage licence (RM120,000.00 per application) and (iv) initiation of storage site closure (RM120,000.00 per application)." Operators must "monitor the storage complex and the surrounding environment, and prepare a monitoring plan; carry out any corrective measures and remediation measures with regard to any leakage or significant irregularity; and submit to the Agency a report of the result of the monitoring" (Rajah & Tann/Wong & Partners summary on Lexology).

Norway (Norwegian Offshore Directorate, formerly NPD): exploitation permit EL001 "Aurora" was awarded in January 2019 for Northern Lights; per Northern Lights JV, "The permit granted is for the injection and storage of 37.5 million tonnes of CO2 from this year and the next 25 years." Norwegian implementation transposes the CCS Directive through Storage Regulations (1517/2014) and Pollution Regulations (931/2004) (Haavind).

Australia OPGGS Act 2006 (Chapter 3, Part 3.2) requires Greenhouse Gas Assessment Permits → Holding Leases → Injection Licences via NOPTA (title administration) and NOPSEMA (safety/environment). Companies must submit Environment Plans (under OPGGS (Environment) Regulations 2023) and well operations management plans before any activity starts (industry.gov.au).

Table-stakes feature comparison:

Capability	CMG GEM	SLB INTERSECT	tNavigator	MRST-co2lab	TOUGH/ECO2N	CarbonLens (current)
3D compositional multi-phase flow	✓ full	✓ full	✓ full	✓ full	✓ full	Screening-only

Capability	CMG GEM	SLB INTERSECT	tNavigator	MRST-co2lab	TOUGH/ECO2N	CarbonLens (current)
Coupled geochemistry (mineral trapping)	✓ GAM	✓	✓	Partial	✓ ECO2N	Solubility only (Duan-Sun)
Geomechanics coupling	✓ stress/strain	✓ 4D geomech	✓	✓	External	Mohr-Coulomb + MAIP (screening)
History matching (assisted)	✓ CMOST	✓	✓	Adjoints	External	Not present
Adjoint optimization / well placement	Partial	✓	✓	✓ AD	×	Not present
Seismic / Petrel integration	Via export	Native	Native	Eclipse-deck import	Limited	Not present
Wellbore hydraulics	CO2LINK	✓	✓	Limited	✓ T2Well	Not present
UQ (P10/P50/P90 ensemble)	✓ CMOST	✓	✓	✓	External	✓ (core strength)
ML-accelerated property prediction	Limited	✓ ML modules	Limited	Limited	×	✓ (MARS IFT)
Browser-native / zero install	×	Delfi cloud	×	×	×	✓ (unique)
LAS file import	Via Builder	Native	Native	Manual	Manual	✓
Jurisdiction-specific permit export	×	×	×	×	×	✓ (unique)

The strategic implication is clear: CarbonLens is **not** a CMG GEM substitute. Its three defensible moats are (1) ML-accelerated property prediction (MARS IFT), (2) browser-native zero-install UX, and (3) jurisdiction-aware permit export — a niche no incumbent occupies.

4. Pricing and Commercial Strategy

Verified pricing benchmarks:

- **CMG (TSX:CMG) FY2025 revenue: CAD \$129.4 million, up 19% YoY** (GlobeNewswire, 22 May 2025). Per the same release, "Total revenue increased by 19% (1% Organic decline and 20% growth from acquisitions) to \$129.4 million; Recurring revenue increased by 13% (1% Organic growth and 12% was growth from acquisitions) to \$86.8 million; Adjusted EBITDA increased by 2% to \$44.0 million; Adjusted EBITDA Margin was 34%." Approximately 600 customers in 60 countries (D&B company profile) → implied blended ARPC ~CAD \$216K/year (note: post-acquisition figure includes BHV/Bluware/SeisWare; pre-acquisition reservoir-simulation-only ARPC was historically ~CAD \$160K/client/year).
- **Petrel: USD \$30,000–\$50,000 per user per year** (Open iT: "With annual costs ranging from \$30,000 to \$50,000 per user, Petrel software licenses are indispensable for petroleum engineers, reservoir engineers, geophysicists, and geologists").
- **tNavigator full module bundle list-price: USD \$3.75 million** (UCSI University MoU, Rock Flow Dynamics: "The company is donating tNavigator™ software worth \$3,750,000...").
- **Rock Flow Dynamics revenue: \$33.8M in 2024, 225 employees** (Latka).
- **SLB INTERSECT** licensed via Delfi platform subscription (proprietary, not publicly disclosed; comparable to Eclipse).
- **TOUGH2/ECO2N, MRST-co2lab, OPM, BOAST**: free (LBNL/SINTEF/OPM/DOE open-source).

Comparable disruption/exit precedents:

- **SimScale (browser CAE)**: ~\$60M cumulative funding; "In September 2024, the company announced that it had 600,000 registered users on its platform" (Wikipedia citing SimScale September 2024 announcement); Insight Partners-led Series C €27M (January 2020) + €25M extension (October 2021, Draper Esprit + Insight).
- **Seequent (3D geology, Leapfrog)**: acquired by Bentley Systems. Per Bentley Systems investor release of 11 March 2021, "Bentley Systems Enters into ~\$1.05 Billion Agreement to Acquire Seequent, Global Leader in 3D Modeling Software for the Geosciences." Per Bentley's SEC 10-K (FY2021), "On June 17, 2021, the Company completed the acquisition of Seequent... for \$883,336 [thousand] in cash, net of cash acquired, plus 3,141,342 shares of the Company's Class B Common Stock." This is the most relevant exit precedent — a geoscience-focused software company built on a single category (3D implicit geological modeling) exiting at >\$1B to a CAD/CAE incumbent.
- **Resoptima** (Norwegian reservoir UQ startup): per Halliburton's official press release of 5 June 2023, "Halliburton Company (NYSE: HAL) today announced it acquired Resoptima AS, a leading Norwegian technology company that specializes in data-driven reservoir management... To date, more than 130 active fields globally have benefited from Resoptima's technology, enjoying improved production volume predictions and comprehensive assessments of uncertainties and risks." Purchase price undisclosed.
- **ResFrac** (subsurface simulation): per Banneker Partners press release of 22 April 2026, "Banneker Partners ('Banneker') today announced a platform investment in ResFrac Corporation ('ResFrac'), the developer of the industry's only fully integrated reservoir simulation and hydraulic fracturing platform." Prior total funding \$5.4M per PitchBook; deal terms undisclosed.

Recommended pricing model for CarbonLens:

Tier	Price (USD/year)	Target customer	Features
------	---------------------	--------------------	----------

Tier	Price (USD/year)	Target customer	Features
Academic / Free	\$0	Universities, students	Full functionality, watermarked exports, no SLA
Practitioner	\$4,800/yr (\$400/mo)	Independent consultants, small E&Ps	LAS import, full ML, permit-export PDF, 3D viz
Team	\$24,000/yr (3 seats)	Engineering boutiques, regulators	Multi-user, collaboration, version control
Enterprise	\$80,000–\$150,000/yr	NOCs (PETRONAS), supermajors (Shell, Equinor, Chevron)	SSO, on-prem deployment option, dedicated support, regulatory compliance updates, API for CMG/INTERSECT integration, custom permit templates
Research/National Lab	Negotiated	NETL, LBNL, SINTEF, CO2CRC	Co-development license

This tiering is anchored on (a) Petrel/CMG benchmarks for enterprise comparability and (b) SimScale's freemium-to-enterprise SaaS playbook.

Grants and accelerators to pursue:

- **DOE SBIR/STTR (US)**: Per U.S. DOE/AMMTO's official SBIR page, "Phase I projects explore the feasibility of innovative concepts with awards of up to \$200,000 (depending on the topic) over nine months. Phase II projects are expanded R&D efforts, with awards of up to \$1,100,000 over two years." NETL Carbon Storage R&D solicitations recur annually.
- **EU Horizon Europe / Innovation Fund (CCS calls)**, particularly under the Industrial Carbon Management Strategy.
- **Malaysian MOSTI** technology development grants; **Malaysian Technology Development Corporation (MTDC)** commercialization funds; **MyCCUS Agency** R&D pull-through.
- **PETRONAS Research Fund (PRF)** — UTP is uniquely positioned given PETRONAS ownership.
- **JOGMEC / METI** (Japan) joint R&D under the PETRONAS-JOGMEC-METI Memorandum of Cooperation (cited in Norton Rose Fulbright and AmCham Malaysia briefings).
- **ASEAN CCS R&D platform** (under ASEAN Strategy for Carbon Neutrality).

Details

Journals & publication strategy

Primary targets (ranked by relevance for software-platform CCS PhD): (1) *International Journal of Greenhouse Gas Control* (Elsevier; 2024 IF 6.19 per Resurchify; Q1; editor Samuel Krevor, Imperial; APC USD \$3,900); (2) *Computers & Geosciences* (Elsevier); (3) *Fuel* (Elsevier; broad scope, accepts ML IFT papers — e.g. Shang et al. *Fuel* 2026, 405, 136502, "An explicit machine learning model for brine-gas

interfacial tension prediction: Implications for H₂, CH₄, and CO₂ geo-storage"); (4) *Energy & Fuels* (ACS; published the 2025 SL/ENN/Power Law IFT ensemble paper); (5) *SPE Reservoir Evaluation & Engineering* (industry-credible for Class VI permit-application papers; published "Building an EPA Class VI Permit Application," 2023); (6) *Advances in Water Resources, Transport in Porous Media, Geoenergy Science and Engineering*; (7) *Applied Energy* (techno-economic CCS); (8) *Environmental Science: Advances* (RSC; published a 2025 review of ML for CO₂-brine IFT in saline aquifers).

PETRONAS / Malaysian pitchability — the operational case

PETRONAS' CCS pipeline maps tightly onto CarbonLens's feature set:

- **Kasawari Phase II (Sarawak, SK316, 108 m water depth)**: per NS Energy, project will "contribute to reducing carbon dioxide emissions emitted via flaring by 3.3 million tonnes per annum (mtpa), thereby making it one of the largest offshore CCS projects in the world." Scheduled to start operations late-2025; recent LinkedIn industry posts indicate slip to 2027. EPCIC by MMHE; corrosion prediction software for supercritical CO₂ already procured (NS Energy).
- **Duyong (Peninsular Malaysia)**: Malaysia's first Offshore Assessment Permit, awarded 10 October 2025 to PCCSV+TotalEnergies+Mitsui; FEED phase next (PETRONAS press release, 10 November 2025).
- **Penyu Basin**: JSDA with ADNOC + Storegga (20 August 2025), targeting ≥5 Mtpa CO₂ capture-and-storage capacity by 2030; "geophysical and geomechanical modelling, reservoir simulation and containment research... application of artificial intelligence" explicitly named (Argus Media).
- **Lang Lebah** (PTTEP + PCSB + KUFPEC) and **Kerteh/Kuantan onshore hubs** also in pipeline (Norton Rose Fulbright).
- Malaysia's national target per Norton Rose Fulbright: "the development of three hubs by 2030 (two in Peninsular Malaysia and one in Sarawak), with a combined storage capacity of up to 15 Million Tonnes Per Annum (mtpa) and which is expected to expand to 40–80 Mtpa across three major hubs by 2050."

This is roughly 15–20 Mtpa of permit-application work over 2026–2030 in Malaysia alone, plus cross-border CCS shipping into Malaysia from Japan (JERA), South Korea (K-Line/JAPEX), and Singapore. Every one of these will require a permit application under the CCUS Act 2025 — and at RM 80,000 per assessment permit and RM 120,000 per storage licence, the regulatory documentation burden is non-trivial.

A jurisdiction-aware permit-export tool with embedded MyCCUS-Act-compliant templates has direct, immediate utility for PCCSV and its JV partners.

Technical feasibility & gaps in detail

The browser-native architecture (React Three Fiber, WebGL, WASM) imposes hard limits:

- **Field-scale 3D compositional simulation (>1M cells)** is not feasible client-side. Solution: offload heavy runs to a thin cloud-compute backend (NETL's NRAP Open-IAM is open source and could be the integration point) or partner with CMG Cloud / SLB Delfi.
- **History matching at field scale** requires ensemble methods (MCMC, EnKF, ES-MDA) that need persistent compute. CMG CMOST and tNavigator's assisted-HM are the benchmarks.
- **Geochemistry** (Duan-Sun is solubility-only) needs to be extended to mineral trapping (PHREEQC.js integration) for credible Class VI submissions.
- **Seismic interpretation** — CarbonLens has no Petrel-equivalent for seismic interpretation, fault tracking, or structural modeling. The pragmatic answer is **read-only Petrel/Eclipse-deck import**

rather than building from scratch (MRST already provides open-source Eclipse-deck parsers per SINTEF documentation).

Recommended technical roadmap (24-36 months)

1. **Months 0–6:** Publish MARS IFT validation paper in *Fuel* or *International Journal of Greenhouse Gas Control*. Submit Class VI Area-of-Review benchmark study against NETL Open-IAM and MRST-co2lab. Open-source the MARS model with permissive license to seed academic adoption.
2. **Months 6–12:** Add PHREEQC.js integration for mineral-trapping geochemistry. Add Sleipner / Johansen / Utsira public benchmark cases pre-loaded. Submit *Computers & Geosciences* paper on the browser-native UQ pipeline.
3. **Months 12–18:** Build read-only Eclipse-deck import (port MRST parsers to JS/WASM). Add cloud-compute backend (AWS Batch or Google Cloud Run) for >100K-cell models. Launch paid tier.
4. **Months 18–24:** Partner with one Malaysian CCS project (Duyong or Kasawari) for permit-template validation. Formalize an MOU with PCCSV or a PETRONAS Research Fund grant.
5. **Months 24–36:** SBIR Phase I (or PRF equivalent) for ML-surrogate history matching. First enterprise customer.

Recommendations

Stage-gated decision framework:

Stage 1 — PhD framing decision (next 3 months)

- **Do:** Frame the PhD as "ML surrogate models and uncertainty quantification for CCS site screening, with browser-native operational deployment under the Malaysian CCUS Act 2025 regulatory framework." This gives you three publishable threads (ML model, UQ method, regulatory ontology) and the platform is the artifact, not the contribution.
- **Don't:** Frame it as "I built an app." Software engineering alone does not survive a viva.
- **Threshold to change:** If you cannot get a PETRONAS-aligned supervisor at UTP (or a PRF grant within 6 months), pivot to a UK EngD at Edinburgh/Heriot-Watt with PETRONAS as the industrial sponsor.

Stage 2 — Product roadmap (months 3–18)

- **Do:** Get one peer-reviewed paper out in *Fuel* or *International Journal of Greenhouse Gas Control* within 12 months — this is your single most important credibility signal for both PhD examiners and enterprise customers.
- **Do:** Add PHREEQC.js + Sleipner benchmark + Eclipse-deck import. These three unlock the "credible companion to CMG/INTERSECT" positioning.
- **Don't:** Try to compete head-to-head with CMG GEM on full 3D compositional simulation. You cannot win this in a browser, and trying will burn the brand.
- **Threshold to change:** If by month 18 you have <100 academic users and no LOI from any operator, narrow scope to a single jurisdiction (Malaysia) and pivot to consulting-as-a-service.

Stage 3 — Commercial launch (months 18–36)

- **Do:** Free academic tier + \$4.8K/yr practitioner + \$80–\$150K/yr enterprise. Start with Malaysian-jurisdiction permit export as the unique paid feature.

- **Do:** Apply for DOE SBIR Phase I (if able to partner with a US small business) — up to \$200,000 over nine months per DOE/AMMTO — or MTDC/MOSTI Malaysian commercialization grant. Phase II is up to \$1,100,000 over two years.
- **Do:** Pursue strategic LOI with one of (PCCSV, TotalEnergies CCS, Mitsui, JOGMEC, NETL/LBNL, SINTEF). Even an unpaid co-development MOU is worth more than three customer demos.
- **Threshold to change exit strategy:** If by year 4 you reach \$1–3M ARR, target Series A; if you reach \$5M+ ARR, target strategic acquisition by Bentley, SLB, or Halliburton. Reference outcomes: Seequent acquired by Bentley for ~\$1.05B (March 2021 agreement; June 2021 close at \$883.3M cash + 3.14M Class B shares per Bentley 10-K); Resoptima acquired by Halliburton at undisclosed price (June 2023, used on 130+ fields).

Stage 4 — Long-term defensibility

- **Do:** Maintain the open-source MARS model and permit-export schema as community standards. The Seequent and SimScale precedents both relied on owning a standard category, not on locked-down code.
- **Do:** Publish 4–6 papers across the PhD to anchor the academic moat.
- **Don't:** Take VC money until you have $\geq$ \$1M ARR or a strategic partner LOI; CCS-software TAM is small enough that early dilution will kill optionality.

Caveats

1. **CCS software TAM is small and slow-moving relative to general CAE.** Mordor Intelligence's \$15.46B simulation-software total is dominated by automotive/aerospace; the CCS slice is plausibly \$100–500M today, growing fast but from a low base. A \$50M ARR business is realistic at 7–10 years; a \$500M ARR business is not.
2. **Per-user/year list pricing for CMG GEM, SLB INTERSECT, Halliburton Nexus, and tNavigator is NOT publicly disclosed** by the vendors. The verifiable Petrel benchmark (\$30K–\$50K/user/yr from Open iT) is the most-cited public figure; commonly cited estimates of "\$160K–\$230K/year for CMG GEM" are not directly substantiated in primary sources and should be treated as practitioner-reported estimates rather than vendor list price. CMG's blended ARPC of ~CAD \$216K/client/year is calculated from FY2025 revenue (CAD \$129.4M) $\div$ ~600 clients but includes post-acquisition revenue from BHV/Bluware/SeisWare.
3. **Browser/WebGL/WASM has hard performance ceilings.** Field-scale 3D compositional simulation with >1M cells, full geochemistry, and history-matching ensembles cannot run client-side. CarbonLens must accept being a screening/pre-FEED/permit-prep tool, OR add a server-side compute layer (which compromises the "zero backend" thesis).
4. **Malaysian CCUS Act 2025 does not apply to Sabah/Sarawak** — exactly where Kasawari sits. Sarawak operates under the 2022 Land Code and 2023 Ordinance (Norton Rose Fulbright). This regulatory fragmentation must be handled in the permit-export module.
5. **Some claims in the brief are not fully verified.** "MARS IFT model" is a developer-original framing; the published ML IFT literature is dominated by random-forest, XGBoost, ENN, and Super Learner approaches (e.g. Nait Amar et al., *Energy & Fuels* 2025, $R^2 = 0.9953$; ACS Omega 2024 Bayesian RF, $R^2 = 0.9729$). The novelty of MARS specifically vs. these methods must be defended explicitly in the validation paper.
6. **Bentley/Seequent and Halliburton/Resoptima exit precedents are encouraging but not deterministic.** Both targets had decade-long operational track records before exit (Resoptima

"Launched in 2013" per Halliburton; Seequent founded 2004); CarbonLens is at PoC stage.

7. **Kasawari start-up date is contested.** Multiple PETRONAS and IPTC sources commit to "end of 2025" for first injection, while a 16 January 2025 LinkedIn industry post (Oil & Gas Lang Perdana) flags acceleration plans to 2027. PETRONAS' own first-gas Kasawari milestone occurred in 2024 (Argus, 20 August 2025), but the CCS phase remains the milestone being tracked.
8. **The 59-tool NETL "Rules and Tools Crosswalk"** is the most authoritative US Class VI permitting computational-tools reference, but it was published in May 2022 and does not reflect the 2024–2025 wave of ML-augmented CCS tools (including arguably CarbonLens itself). This is itself a publishable contribution opportunity: an updated Crosswalk inclusive of ML/browser-native tools.
9. **Subagent-supplied figures:** CMG FY2025 revenue (CAD \$129.4M, GlobeNewswire 22 May 2025), Seequent acquisition (\$1.05B announced March 2021 / \$883.3M close June 2021, Bentley investor release + SEC 10-K), SimScale total funding (~\$60M cumulative; 600,000 users September 2024), DOE SBIR Phase I/II ceilings (\$200K / \$1.1M, DOE/AMMTO), Halliburton-Resoptima (5 June 2023, 130+ fields), Banneker-ResFrac (22 April 2026 platform investment), and NETL "59 tools" Rules-and-Tools count (NRAP-TRS-I-001-2022, 31 May 2022) are quoted from named secondary sources verified against their primary press releases or government documents.